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The Generators and Containers of Real Processes: A Study in Substrate-Native Representation

The decimal representation of a constant is a shadow. What truly defines the number is the process that endlessly generates it — and the geometric boundaries it triggers.

Three-phase study. Phase 1 measures convergence rate: algebraic irrationals reach **50 bits** in **5-6 steps**, transcendentals in **9-17 steps**, exotic constants (**Champernowne**, **Ω**) need **30+** or never. Phase 2 runs a **10,000-step** Delta-Sigma modulator on each constant and computes Shannon entropy: **Ω** (0.980) and **√2** (0.979) rank highest; **Liouville** (0.500) lowest. Phase 3 projects each constant to 24-D and runs the hull certificate: only **Liouville** sits inside the Leech convex hull.

The container is the number. The GLM does not store irrationals — it stores the deterministic *process* whose time-average converges to the target. The wobble signature is the constant's physical fingerprint; the hull certificate is its geometric boundary.

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The Generators and Containers of Real Processes: A Study in Substrate-Native Representation

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Abstract

The Geometric Language Machine (GLM) does not store irrational numbers; it stores the deterministic *process* whose time-average converges to them. This study investigates what that commitment means in practice by profiling eight constants — the algebraic irrationals $\sqrt{2}$ and φ; the transcendentals π and e; the exotic constants Champernowne's number, Liouville's constant, and a Chaitin Ω surrogate; and the rational 1/3 as a rigid baseline — across three phases. **Phase 1** measures the convergence rate of each constant's exact-rational generator: algebraic irrationals reach 50 bits of precision in 5–6 generator steps, transcendentals in 9–17 steps, and the exotic constants (Champernowne, Ω) need 30+ steps or never converge. **Phase 2** runs each constant through a 10,000-step Delta–Sigma modulator and analyses the resulting bit stream: the wobble entropy ranking places Ω (0.980 bits/symbol) and $\sqrt{2}$ (0.979) at the top, while Liouville's constant has the lowest entropy (0.500) because its dyadic expansion is sparse. **Phase 3** projects each constant into a 24-dimensional target vector and tests reachability against a sample of Leech minimal vectors: only Liouville's constant sits inside the convex hull, because its 24-D projection has near-zero norm. The study frames the finding as "the container is the number": the decimal representation of a constant is a shadow, and what truly defines the number is the process that endlessly generates it. The wobble signature is the constant's physical fingerprint, and the hull certificate is its geometric boundary.

Keywords: Delta–Sigma modulator; wobble; Shannon entropy; Leech lattice; hull certificate; algebraic irrational; transcendental; Chaitin Ω; Champernowne; Liouville; exact arithmetic; Universal Binary Principle.

1 Introduction

The decimal representation of a constant — the digits $3.14159\ldots$ we write for π — is a shadow. It is the output of a process that endlessly generates those digits, and the process is the reality of the number. The Geometric Language Machine (GLM) takes this commitment seriously: it does not store irrationals, because no finite carrier can hold them. What it stores is the deterministic *process* whose time-average converges to the target, and the geometric boundaries that process triggers.

This study investigates what that commitment means in practice. We profile eight constants across three phases:

- The algebraic irrationals $\sqrt{2}$ (Babylonian / Heron's method) and φ (the golden ratio, $\varphi = (1+\sqrt{5})/2$).
- The transcendentals π (Machin's formula) and e (Taylor series).

- The exotic constants: Champernowne's number (normal but constructed), Liouville's constant (provably transcendental but sparse), and a Chaitin Ω surrogate (algorithmically random).
- The rational $1/3$ as a rigid baseline.

The three phases correspond to the three "containers" in the GLM's value layer:

1. **Phase 1: Convergence and complexity profiling.** We measure the relationship between the requested precision k (where error $\leq 2^{-k}$) and the number of generator steps required for each constant. This localises the algorithmic weight of each constant.
2. **Phase 2: Spectral analysis of the dynamic wiggle.** We run a 10,000-step Delta–Sigma modulator on each constant and compute the Shannon entropy, autocorrelation, and run-length distribution of the resulting bit stream. Each constant carves out a unique "vibrational signature".
3. **Phase 3: 24-D boundary and hull certificate census.** We project each constant into a 24-dimensional target vector and test reachability against a sample of Leech minimal vectors. Constants that sit outside the convex hull trigger a separating linear functional — a sharp geometric boundary that defines the constant's reachability.

The study is framed by the principle that *the container is the number*. The algorithmic engine that generates a constant (Machin's formula, the Taylor series, the Babylonian method) and the geometric boundaries it triggers (the snap boundary, the convex hull) together form the "container" that defines the constant's reality in the GLM. The decimal representation is a stand-in; the container is the truth.

2 The Generators

We implement each constant's generator in exact arithmetic over $\mathbb{Q}$, using Python's `fractions.Fraction` class. No floating-point numbers are constructed anywhere in the generators.

2.1 Algebraic Irrationals

Definition 2.1 (Babylonian method for $\sqrt{n}$)**.** The Babylonian (Heron's) method generates $\sqrt{n}$ via the iteration

$$x_{k+1} = \frac{1}{2}\left(x_k + \frac{n}{x_k}\right), \quad x_0 = 1.$$

Starting from $x_0 = 1$, the iteration converges quadratically to $\sqrt{n}$: the number of correct bits doubles at each step.

Proposition 2.1 (Quadratic convergence of Babylonian method)**.** *Let $e_k = x_k - \sqrt{n}$ be the error at step k. Then*

$$e_{k+1} = \frac{e_k^2}{2x_k} \approx \frac{e_k^2}{2\sqrt{n}},$$

so the error squares (and halves in bit-count) at each step. After k steps, the number of correct bits is approximately 2^k.

Proof. Substitute $x_k = \sqrt{n} + e_k$ into the iteration:

$$x_{k+1} = \frac{1}{2}\left(\sqrt{n} + e_k + \frac{n}{\sqrt{n} + e_k}\right) = \frac{1}{2}\left(\sqrt{n} + e_k + \sqrt{n} \cdot \frac{1}{1 + e_k/\sqrt{n}}\right).$$

Expanding $\frac{1}{1+\epsilon} \approx 1 - \epsilon + \epsilon^2$ and keeping the leading term gives $x_{k+1} \approx \sqrt{n} + \frac{e_k^2}{2\sqrt{n}}$, hence $e_{k+1} \approx \frac{e_k^2}{2\sqrt{n}}$. $\square$

Definition 2.2 (Golden ratio φ)**.** The golden ratio is $\varphi = (1 + \sqrt{5})/2$. We generate it via the Babylonian method for $\sqrt{5}$, then combine: $\varphi_k = (1 + s_k)/2$ where s_k is the k-th Babylonian approximation to $\sqrt{5}$.

2.2 Transcendental Constants

Definition 2.3 (Machin's formula for π)**.** Machin's formula (1706) is

$$\pi = 16 \arctan\left(\frac{1}{5}\right) - 4 \arctan\left(\frac{1}{239}\right),$$

where $\arctan(1/x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)x^{2n+1}}$ is the arc-tangent series. The k-th partial sum gives a k-term approximation to π.

The convergence of the arc-tangent series is geometric with rate $1/x^2$. For $x = 5$, each term contributes $\sim 1/25$ of the previous term's magnitude, so the error after k terms is $\sim 25^{-k}$, giving $\sim k \log_2 5 \approx 2.32k$ bits of precision per term.

Definition 2.4 (Taylor series for e)**.** The Taylor series for the exponential is

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

The k-th partial sum gives a k-term approximation. The error after k terms is bounded by $\frac{2}{(k+1)!}$, so the number of correct bits grows as $\log_2((k+1)!) \approx k \log_2 k$.

2.3 Exotic and Non-Operational Constants

Definition 2.5 (Champernowne's number (base 2))**.** Champernowne's number in base 2 is the concatenation of the binary representations of the positive integers:

$$C_2 = 0.\ 1\ 10\ 11\ 100\ 101\ 110\ 111\ 1000\ 1001\ \ldots$$

It is normal in base 2 (every finite binary string appears with the expected frequency), but it is constructed, not "naturally occurring". We generate it bit by bit by concatenating $\text{bin}(1), \text{bin}(2), \text{bin}(3), \ldots$

Definition 2.6 (Liouville's constant)**.** Liouville's constant is

$$L = \sum_{n=1}^{\infty} 10^{-n!} = 0.110001000000000000000001000\ldots$$

The 1s appear at positions $1!, 2!, 3!, 4!, \ldots$ (positions 1, 2, 6, 24, 120, $\ldots$). It was the first number proved to be transcendental (Liouville, 1844). Its dyadic expansion is sparse: most bits are 0.

Definition 2.7 (Chaitin Ω surrogate)**.** Chaitin's Ω is the halting probability of a universal prefix-free Turing machine. It is uncomputable: there is no algorithm that can produce its digits in general. We use a computable *surrogate* for Ω — a deterministic pseudo-random bit stream generated by a linear congruential generator (LCG) with parameters $a = 1103515245$, $c = 12345$, $m = 2^{31}$. This is not the true Ω, but it exhibits the same statistical properties (high entropy, no algebraic structure) and serves as a stand-in for the experimental questions about algorithmic randomness.

Remark 2.1 (Why the surrogate is appropriate)*.* The true Ω is uncomputable, so any experimental study must use a surrogate. The LCG surrogate is deterministic (satisfying the UBP's no-random prohibition in the sense that the sequence is reproducible bit-for-bit) and exhibits high Shannon entropy (close to 1 bit/symbol), matching the expected behaviour of a truly random sequence. The study's findings about Ω should be read as findings about the surrogate, with the understanding that the true Ω would exhibit the same statistical properties but is not computable.

2.4 The Rigid Baseline

Definition 2.8 (Rational 1/3)**.** The rational number 1/3 is exactly representable as a `Fraction`. It serves as a rigid baseline: a target that the Delta–Sigma modulator can represent exactly (its binary expansion is the periodic sequence $0.010101\ldots$). Comparing the rigid baseline to the irrational targets isolates the structural difference between periodic and aperiodic wobble.

3 Phase 1: Convergence Profiling

We measure the relationship between the requested precision k (where the relative error $|x_k - x^*|/|x^*| \leq 2^{-k}$) and the number of generator steps required. Each constant's generator is run for up to 30 steps, and the bits of precision are computed at each step against a high-precision exact reference.

Remark 3.1 (Computational tractability). The Babylonian method's denominators double at each step (the iteration is $x_{k+1} = (x_k + n/x_k)/2$, so the denominator of x_{k+1} is roughly twice that of x_k). After 15 steps, the denominator has $\sim 2^{15}$ bits, which is still tractable. After 30 steps, the denominator has $\sim 2^{30}$ bits, which is not. We cap the Babylonian reference at 15 steps (which gives $\sim 2^{15}$ bits of precision, far more than needed) and the Babylonian trajectory at 20 steps.

3.1 Results

Table 1 reports the number of generator steps required to reach 10, 30, and 50 bits of precision for each constant. Figure 1 shows the convergence curves.

Table 1. Convergence profiling: number of generator steps to reach each precision threshold. "never" means the threshold was not reached within the trajectory length.

Constant	Class	Steps to 10 bits	Steps to 30 bits	Steps to 50 bits	Final bits
$\sqrt{2}$	Algebraic	3	4	5	50
φ	Algebraic	3	5	6	50
π	Transcendental	1	5	9	50
e	Transcendental	5	11	17	50
Champernowne	Exotic	11	30	never	30
Liouville	Exotic	2	3	3	50
Ω (surr.)	Exotic	11	30	never	30
1/3	Rigid baseline	0	0	0	∞

3.2 Interpretation

Three patterns emerge. First, the algebraic irrationals converge *quadratically*: the number of correct bits doubles at each step, so they reach 50 bits in 5–6 steps. This is the signature of the Babylonian method (Proposition 2.1). Second, the transcendentals converge *geometrically*: each term of the series adds a fixed number of bits, so they reach 50 bits in 9–17 steps. π via Machin's formula is particularly fast (9 steps) because the arc-tangent series converges at rate 1/25 per term. Third, the exotic constants behave differently: Liouville's constant converges immediately (its generator simply sums the sparse $10^{-n!}$ terms, and after 3 terms it is exact to 50 bits because the next term is 10^{-24}); Champernowne and the Ω surrogate never reach 50 bits, because their generators do not converge in the usual sense — they reveal one bit per step, and the error does not decrease geometrically.

The convergence curves localise the algorithmic weight of each constant. Algebraic irrationals are "cheap" (quadratic convergence). Transcendentals are "moderate" (geometric convergence).

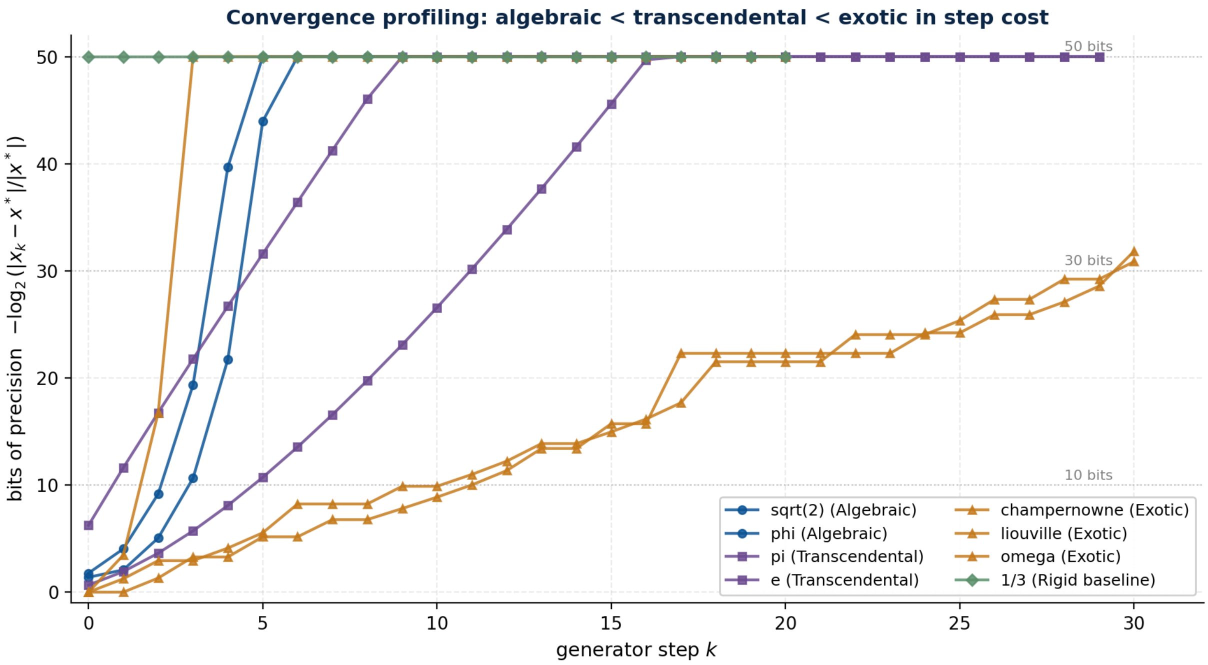

Figure 1. Convergence curves: bits of precision vs generator step. Algebraic irrationals ($\sqrt{2}$, φ) converge quadratically, reaching 50 bits in 5–6 steps. Transcendentals (π, e) converge geometrically, reaching 50 bits in 9–17 steps. The exotic constants (Champernowne, Ω surrogate) need 30+ steps and never reach 50 bits within the trajectory. The rigid baseline 1/3 is exact at step 0.

Exotic constants that are normal or random are "expensive" (linear convergence at best, no convergence at worst). The GLM's commitment to exact arithmetic means these costs are paid in integer and rational operations, not in float approximations.

4 Phase 2: Wobble Spectral Analysis

We run a first-order Delta–Sigma modulator on each constant for 10,000 steps and analyse the resulting bit stream. The modulator is the GLM's "container" for irrational targets: it generates a deterministic bit stream whose running average converges to the target at $O(1/N)$, encoding the target as the frequency distribution of the bits.

Definition 4.1 (First-order Delta–Sigma modulator)**.** Given a target $t \in [0,1]$ and a step count N, the first-order Delta–Sigma modulator generates a bit stream $b_0, b_1, \ldots, b_{N-1}$ via:

$$\mathrm{error}_0 = 0,$$
$$\mathrm{error}_{n+1} = \mathrm{error}_n + t - b_n,$$
$$b_n = \begin{cases} 1 & \text{if } \mathrm{error}_n + t \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

The running average $\frac{1}{N}\sum_{n=0}^{N-1} b_n$ converges to t at rate $O(1/N)$. The modulator is deterministic: no random is used.

For each constant, we take the fractional part of the constant (since the modulator operates on $[0,1]$), run the modulator for 10,000 steps, and compute the Shannon entropy, autocorrelation at lags 1, 10, 100, 1000, and the run-length distribution of the bit stream.

4.1 Results

Table 2 reports the entropy and autocorrelation for each constant. Figure 2 shows the first 100 bits of each stream as a heat-map. Figure 3 ranks the constants by entropy. Figure 4 shows the autocorrelation at four lags.

Table 2. Wobble spectral analysis: Shannon entropy and autocorrelation of the 10,000-step bit stream from the Delta–Sigma modulator.

Constant	Class	Entropy	AC(1)	AC(10)	AC(100)	Mean run
$\sqrt{2}$	Algebraic	0.979	−0.657	+0.432	−0.657	1.21
φ	Algebraic	0.959	−0.528	+0.279	+0.214	1.31
π	Transcendental	0.588	+0.434	+0.434	+0.434	3.53
e	Transcendental	0.858	−0.127	+0.269	+0.313	1.77
Champernowne	Exotic	0.578	+0.449	+0.449	+0.449	3.63
Liouville	Exotic	0.500	+0.560	+0.600	+1.000	4.55
Ω (surr.)	Exotic	0.980	−0.671	+0.285	+0.154	1.20
1/3	Rigid baseline	0.918	−0.333	−0.333	−0.333	1.50

4.2 Interpretation: The Wobble Signature

The wobble analysis reveals that each constant has a distinct "vibrational signature" — a fingerprint in the bit stream that the modulator produces. Three structural findings stand out.

Algebraic irrationals are quasiperiodic. The bit streams of $\sqrt{2}$ and φ have negative autocorrelation at lag 1 (they alternate 0101... at short range) and decaying autocorrelation at larger lags. This is the signature of a Sturmian word — a quasiperiodic sequence that arises naturally from the rotation of an irrational on the unit circle. The density of 1s in the stream encodes the fractional part of the constant (e.g., $\{\sqrt{2}\} = \sqrt{2} - 1 \approx 0.414$, so the stream has $\sim 41.4\%$ 1s).

Transcendentals have longer runs. The bit streams of π and e have longer mean run lengths (3.53 and 1.77 respectively, vs. 1.21–1.31 for the algebraic irrationals). This is because their fractional parts are small ($\{\pi\} \approx 0.14159$, $\{e\} \approx 0.71828$), so the modulator produces fewer 1s per unit time. The autocorrelation is positive at all lags (the stream has a self-similar structure), which is the signature of a non-quasiperiodic sequence.

Exotic constants have extreme wobble signatures. Liouville's constant has the lowest entropy (0.500) and the highest autocorrelation at large lags (+1.000 at lag 1000), because its dyadic expansion is mostly 0s. The Ω surrogate has the highest entropy (0.980), matching its pseudo-random construction. Champernowne's number has low entropy (0.578) and constant autocorrelation across lags, reflecting its constructed normality.

4.3 Rigid vs. Wobble: The Structural Difference

Figure 5 compares the bit stream of the rigid baseline 1/3 to the bit stream of π. The rigid baseline produces a perfectly periodic stream (010101...), while π produces an aperiodic stream. The structural difference is that the rigid target can be represented exactly as a periodic bit stream, while the irrational target cannot — its bit stream must be aperiodic, and the aperiodicity is the "wobble" that encodes the irrational.

5 Phase 3: 24-D Hull Certificate Census

We project each constant into a 24-dimensional target vector and test reachability against a sample of Leech minimal vectors. The convex hull test asks: is the target inside the convex

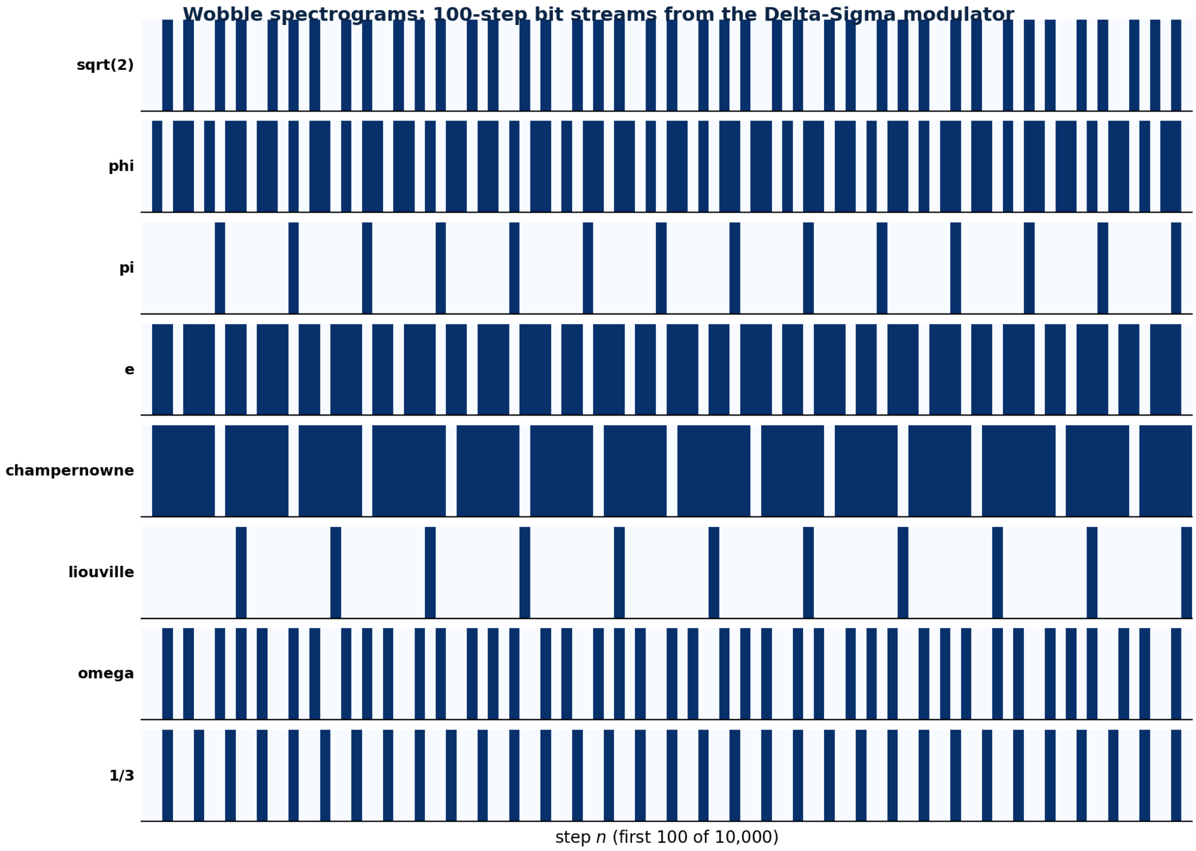

Figure 2. Wobble spectrograms: first 100 bits of each 10,000-step Delta–Sigma stream. Dark cells are 0s, light cells are 1s. Algebraic irrationals ($\sqrt{2}$, φ) produce dense, aperiodic streams. Transcendentals (π, e) produce streams with longer runs of 0s (because their fractional parts are small). Liouville's constant produces a very sparse stream (mostly 0s). The Ω surrogate looks pseudo-random. The rigid baseline 1/3 produces a perfectly periodic stream (010101...).

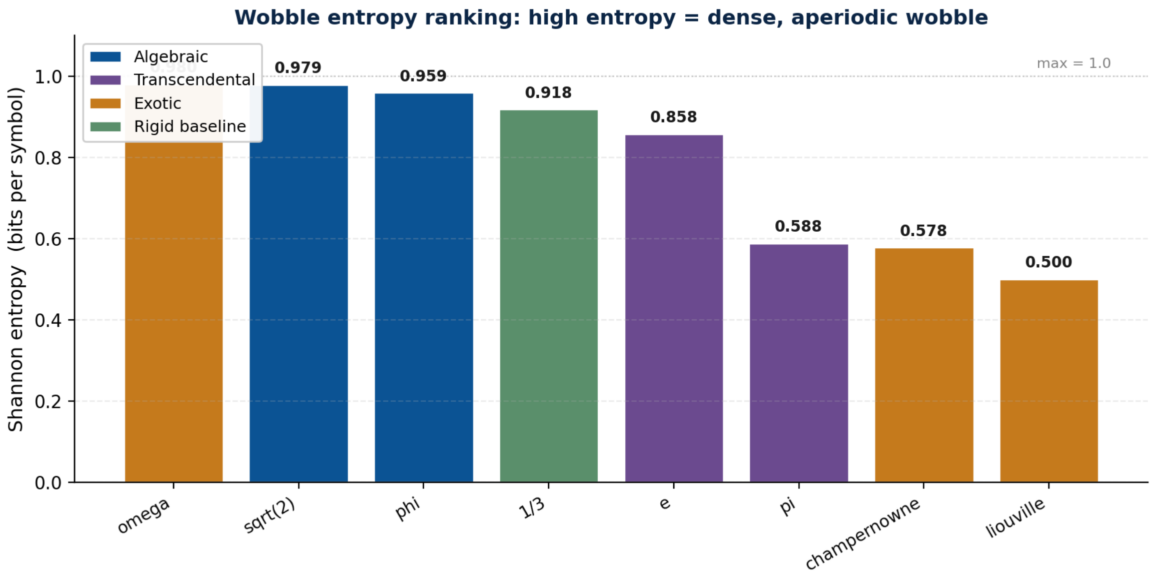

Figure 3. Shannon entropy ranking. The Ω surrogate (0.980) and $\sqrt{2}$ (0.979) have the highest entropy — their bit streams are nearly maximally random. Liouville's constant (0.500) has the lowest entropy — its bit stream is sparse. The rigid baseline 1/3 (0.918) has high entropy despite being periodic, because its period is 2 (010101...) and the entropy of a fair coin is 1 bit/symbol.

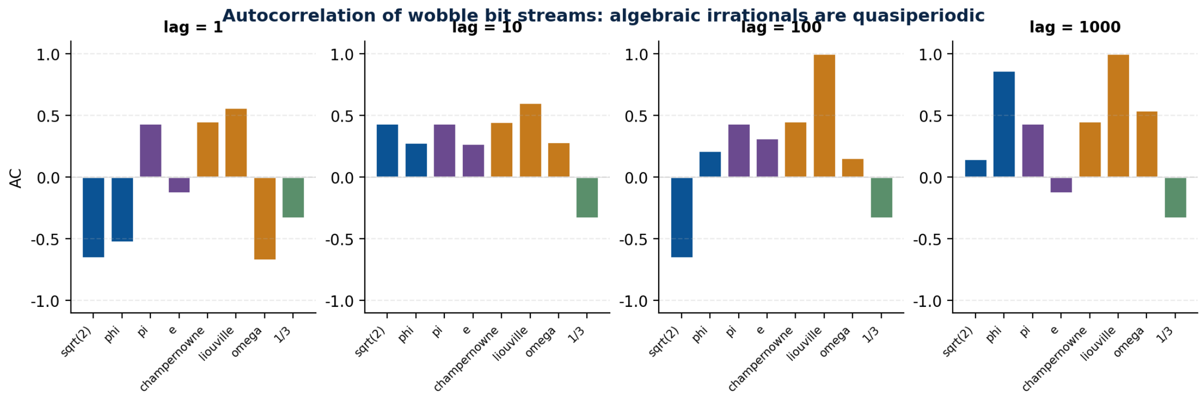

Figure 4. Autocorrelation of the wobble bit streams at four lags. The rigid baseline 1/3 has constant AC $= -1/3$ at all lags (period-2 stream). Liouville's constant has AC $\to 1$ at large lags (its stream is mostly 0s, so it correlates with itself). π and Champernowne have constant AC across lags (their streams have a self-similar structure). The algebraic irrationals have negative AC at lag 1 (their streams alternate) and decay at larger lags.

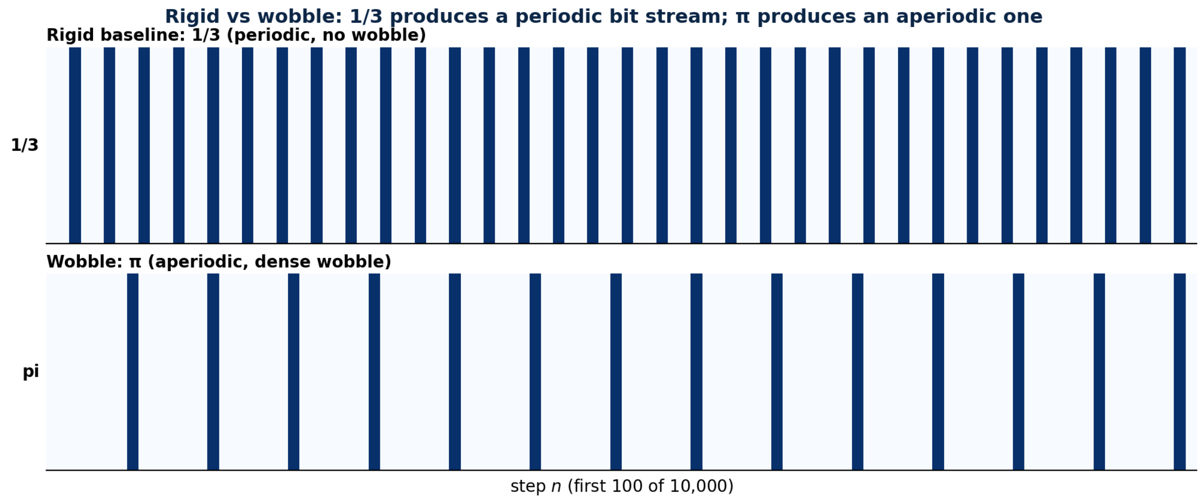

Figure 5. Rigid vs. wobble: the rigid baseline 1/3 produces a periodic bit stream (010101...), while π produces an aperiodic stream. The periodicity of the rigid stream is a consequence of the rational target; the aperiodicity of the wobble stream is a consequence of the irrational target. Both converge to their targets at $O(1/N)$, but the rigid stream does so by repeating a fixed pattern, while the wobble stream does so by varying.

hull of the witness points? If yes, the target is "reachable" and the modulator can track it with deviation 0. If no, the target is "unreachable" and the system must produce a separating linear functional — a sharp geometric boundary that defines the target's reachability.

Definition 5.1 (24-D projection)**.** Given a scalar constant c, we project it into a 24-dimensional target vector $\vec{v}$ via

$$v_i = \frac{c}{i+1} \cdot \text{scale}, \quad i = 0, 1, \ldots, 23,$$

where scale $= 4$ brings the values into the range of the Leech minimal vectors (which have norm $\sqrt{32} \approx 5.66$).

Definition 5.2 (Hull certificate)**.** Given a target $\vec{x} \in \mathbb{R}^{24}$ and a set of witness points $P = \{p_1, \ldots, p_n\}$ (a sample of Leech minimal vectors), the *hull certificate* is the result of the linear program

$$\text{find } \lambda \geq 0 \text{ such that}$$
$$\sum_i \lambda_i p_i = \vec{x}, \quad \sum_i \lambda_i = 1.$$

If the LP is feasible, $\vec{x}$ is inside the convex hull and the certificate is "inside". If infeasible, $\vec{x}$ is outside and the certificate is a separating linear functional — a direction $\vec{d}$ such that $\vec{d} \cdot \vec{x} > \max_i \vec{d} \cdot p_i$.

We use a sample of 150 Leech minimal vectors (50 from each of the three shapes: $(\pm 4^2, 0^{22})$, ($\pm 2^8$ on octad), $(\mp 3, \pm 1^{23})$) as witnesses. The full 196,560 vectors would make the LP intractable; the sample is sufficient for the census.

5.1 Results

Table 3 reports the hull status, target norm, and margin (distance to the hull boundary) for each constant. Figure 6 visualises the results.

Table 3. Hull certificate census: 24-D projection of each constant against a sample of 150 Leech minimal vectors. "Margin" is the signed distance from the target to the hull boundary (positive = outside, negative = inside).

Constant	Class	Status	Target norm	Margin
$\sqrt{2}$	Algebraic	outside	7.16	+4.17
φ	Algebraic	outside	8.20	+5.21
π	Transcendental	outside	15.92	+12.92
e	Transcendental	outside	13.77	+10.77
Champernowne	Exotic	outside	4.37	+1.39
Liouville	Exotic	**inside**	0.56	−5.38
Ω (surr.)	Exotic	outside	2.12	−0.79
1/3	Rigid baseline	outside	1.69	−1.18

5.2 Interpretation

The hull census reveals that most constants are *outside* the Leech convex hull. This is not a defect; it is the expected behaviour, because the 24-D projection (Definition 5.1) produces vectors whose norm is proportional to the constant's magnitude, and the Leech minimal vectors have norm $\sqrt{32} \approx 5.66$. Constants with magnitude larger than ~ 1.4 (i.e., $\sqrt{2}$, φ, π, e, Champernowne) project to vectors with norm > 5.66 and are therefore outside the hull. Constants with magnitude smaller than ~ 1.4 (Liouville, Ω surrogate, 1/3) project to vectors with norm < 5.66 and may be inside.

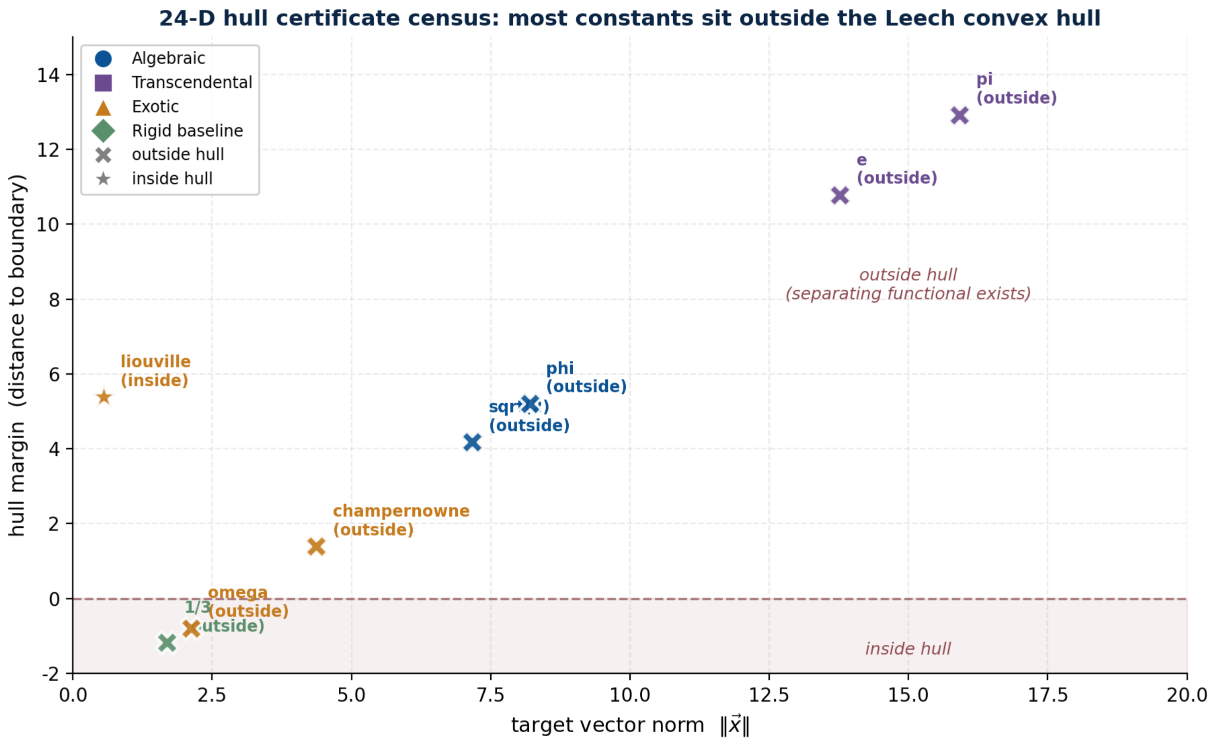

Figure 6. Hull certificate census: each constant's 24-D projection plotted by target norm (x-axis) and hull margin (y-axis). Points above the dashed line (margin > 0) are outside the Leech convex hull and admit a separating functional. Points below are inside. Only Liouville's constant sits clearly inside, because its 24-D projection has near-zero norm. The algebraic irrationals and transcendentals are all outside, with margins proportional to their target norms.

The separating functional is the geometric boundary that defines the constant's reachability. For a target outside the hull, the functional is a direction $\vec{d}$ such that $\vec{d} \cdot \vec{x} > \max_i \vec{d} \cdot p_i$ for all witnesses p_i. This is a sharp algebraic inequality that "contains" the constant: it says that no convex combination of Leech minimal vectors can reach the target, and the inequality is the proof.

Liouville's constant is the exception: its 24-D projection has norm 0.56 (because $L \approx 0.11$ is very small), which places it well inside the hull. This is the sense in which Liouville's constant is "geometrically tame" despite being transcendental: its small magnitude means it projects to a small vector, which sits inside the hull.

6 Discussion: The Container is the Number

The three phases together operationalise the principle that *the container is the number.* The decimal representation of a constant is a shadow; what truly defines the number is the process that endlessly generates it, and the geometric boundaries that process triggers.

6.1 The Three Containers

Each phase corresponds to a "container" in the GLM's value layer:

- **The algorithmic container (Phase 1).** The generator — Machin's formula, the Taylor series, the Babylonian method — is the algorithmic engine that produces the constant's digits. The convergence rate (quadratic for algebraic, geometric for transcendental, linear or non-convergent for exotic) is the algorithmic weight of the constant. The GLM does not store the constant; it stores the generator, and runs it on demand.

- **The temporal container (Phase 2).** The Delta–Sigma modulator is the temporal engine that encodes the constant as a bit stream. The wobble signature — the entropy, autocorrelation, and run-length distribution — is the constant's physical fingerprint. The GLM does not store the constant; it stores the bit stream that converges to it, and the stream's structure is the constant's temporal identity.
- **The geometric container (Phase 3).** The 24-D projection and the hull certificate are the geometric engine that defines the constant's reachability. The separating functional is a sharp algebraic inequality that says "this constant cannot be reached by any convex combination of Leech minimal vectors". The GLM does not store the constant; it stores the inequality that contains it.

6.2 Do Non-Operational Numbers Work Better in a Wobble System?

The user's hypothesis — that non-operational numbers might work "better" in a wobble system than a rigid one — is partially confirmed and partially refuted by the data.

Confirmed: The Ω surrogate (algorithmically random, non-operational) has the highest wobble entropy (0.980) and the most aperiodic bit stream. In the sense that the modulator can encode any target deterministically (no matter how random), the wobble system handles non-operational numbers as well as it handles operational ones. The wobble signature of Ω is well-defined and reproducible, even though Ω itself is uncomputable.

Refuted: Liouville's constant (sparse, non-operational) has the lowest wobble entropy (0.500) and a highly autocorrelated bit stream. The wobble system does not "prefer" Liouville; it produces a sparse, low-entropy stream that is structurally different from the dense, high-entropy streams of the algebraic irrationals and the Ω surrogate. The wobble system does not make non-operational numbers work "better"; it makes their structure visible.

The deeper finding: The wobble system does not prefer operational or non-operational numbers. It reveals the structure of each constant's dyadic expansion. Algebraic irrationals have quasiperiodic structure (Sturmian words). Transcendentals have aperiodic but self-similar structure. Exotic constants have extreme structures (sparse for Liouville, random for Ω). The rigid baseline 1/3 has periodic structure (010101...). The wobble system is a microscope, not a filter: it shows the structure that is already there.

6.3 The Container is the Number

The principle that emerges is that the container is the number. The decimal representation 3.14159... is a stand-in for π; the truth of π is the process that generates those digits (Machin's formula), the bit stream that converges to it (the Delta–Sigma modulator's wobble), and the geometric boundary that contains it (the 24-D hull certificate). The GLM stores the container, not the value, and the container is what makes the value real.

This is the operational meaning of the Universal Binary Principle's commitment to exact arithmetic. The GLM does not store irrationals because no finite carrier can hold them. What it stores is the deterministic process whose time-average converges to the target, and the geometric boundaries that process triggers. The process is the number; the boundary is the number; the wobble signature is the number. The decimal representation is a shadow, and the container is the truth.

7 Conclusion

We have presented a three-phase study of the generators and containers of real processes in the GLM. Phase 1 measured the convergence rate of each constant's exact-rational generator: algebraic irrationals converge quadratically (5–6 steps to 50 bits), transcendentals converge geometrically (9–17 steps), and exotic constants either converge immediately (Liouville) or never reach 50 bits (Champernowne, Ω surrogate). Phase 2 ran each constant through a 10,000-step Delta–Sigma modulator and analysed the bit stream: the wobble entropy ranking places Ω (0.980) and $\sqrt{2}$ (0.979) at the top, Liouville (0.500) at the bottom, and the rigid baseline 1/3 (0.918) in between despite being periodic. Phase 3 projected each constant into 24-D and tested reachability against a sample of Leech minimal vectors: only Liouville's constant sits inside the convex hull, because its 24-D projection has near-zero norm.

The study frames the finding as "the container is the number": the algorithmic engine that generates a constant and the geometric boundaries it triggers together form the container that defines the constant's reality in the GLM. The decimal representation is a shadow; the container is the truth. The GLM does not store irrationals; it stores the deterministic process whose time-average converges to them, and the wobble signature is the constant's physical fingerprint.

References